

1. What is the input() function in Python used for?

To take the input from the user. By default it'll take input as string.

```
In [1]: name = input("Enter your name")
        print(f"Welcome {name}")
```

Welcome Roja

2. How can you accept an integer as input from the user using input()?

```
In [3]: age = int(input("Enter your age"))
        print(f"Your age {age}")
```

Your age 4

3. How do you accept a float input from the user?

```
In [9]: temp = float(input("Enter your area temperature"))
        print(f"Your area temperature {temp:.2f}")
```

Your area temperature 123.68

4. How can you take multiple space-separated values as input?

```
In [19]: multi = input("Enter Multiple Space Separated Value").split()

        for i, val in enumerate(multi):
            print(f"Entered input {i+1}: {val}")
```

Entered input 1: roja
Entered input 2: rani
Entered input 3: repalle
Entered input 4: r

5. How do you check if a number entered by the user is positive, negative, or zero?

```
In [23]: number = int(input("Enter a number"))

        if number < 0:
            print(f"Given number {number} is negative")
        elif number > 0:
            print(f"Given number {number} is positive")
        else:
            print(f"Given number {number} is zero")
```

Given number 0 is zero

6. How do you convert user input to a list of integers?

```
In [52]: list_num = [int(x) for x in input("Enter a number").split()]
list_num
```

Out[52]: [12, 14]

7. How do you accept a string input and print it in uppercase?

```
In [53]: upper_str = input("Enter string").upper()
print(f"Given string is {upper_str}")
```

Given string is ROJA

8. Write a Python program that accepts a string and prints the number of vowels in it.

```
In [56]: string = input("Enter string").lower()
vowels = ['a','e','i','o','u']
count = 0

for i in string:
    if i in vowels:
        count+=1
print(f"Number of vowels in the given string {string} is {count}")
```

Number of vowels in the given string roja is 2

9. Write a program that takes a number as input and checks if it is even or odd.

```
In [60]: number = int(input("Enter a number"))

if number%2 == 0 :
    print(f"Given number {number} is even number")
else:
    print(f"Given number {number} is odd number")
```

Given number 12 is even number

10. How would you check if a string is a palindrome using input()

```
In [70]: p_string = input("Enter a palindrom string")
string_r = ''
for i in p_string[::-1]:
    string_r+=i
if p_string == string_r:
    print(f"Given string {p_string} is a palindrome")
else:
    print(f"Given string {p_string} is not a palindrome")
```

Given string madam is a palindrome

11. Write a program that takes a number as input and prints its square.

```
In [74]: number = int(input("Enter a number"))
print(f"Square of a given number {number} is {number*number}")
```

Square of a given number 4 is 16

12. Write a program that asks for a number and prints whether it is divisible by 3.

```
In [77]: number = int(input("Enter a number"))
if number%3 == 0 :
    print(f"Given number {number} is divisible by 3")
else:
    print(f"Given number {number} is not divisible by 3")
```

Given number 5 is not divisible by 3

13. How would you check if a number is divisible by both 3 and 7?

```
In [80]: number = int(input("Enter a number"))
if number%3 == 0 & number%7 == 0:
    print(f"Given number {number} is divisible by 3 and 7")
else:
    print(f"Given number {number} is not divisible by 3 and 7")
```

Given number 20 is not divisible by 3 and 7

14. How do you accept a list of comma-separated values as input?

```
In [81]: comma_data = input("Enter Multiple Space Separated Value").split(',')
for i,val in enumerate(comma_data):
    print(f"Entered input {i+1}: {val}")
```

Entered input 1: 12

Entered input 2: 13

15. Write a Python program that takes two numbers as input and prints their product.

```
In [82]: number_1 = int(input("Enter the 1st number"))
number_2 = int(input("Enter the 2nd number"))

print(f"Product of 2 numbers {number_1} and {number_2} is {number_1*number_2}")
```

Product of 2 numbers 2 and 3 is 6

16. Write a program that checks if the input number is a prime number.

```
In [83]: number_prime = int(input("Enter the number"))

if number%2 == 0 & number%3 == 0:
    print(f"Given number {number_prime} is not prime number")
else:
    print(f"Given number {number_prime} is prime number")
```

Given number 2 is not prime number

17. How can you accept a boolean value (True/False) from the user?

```
In [88]: bool_val = bool(input("Enter the boolean value"))
print(f"Given boolean value {bool_val}")
```

Given boolean value True

18. Write a program that accepts a string and prints the reverse of that string.

```
In [87]: string =input("Enter a string")
print(f"Given string {string} & reverse of that string is {string[::-1]}")
```

Given string roja & reverse of that string is ajor

19. Write a program that asks for a user's name and age and prints a message.

```
In [91]: name = input("Enter your name")
age = int(input("Enter your age"))
print(f"Welcome {name} and Your age is {age}")
```

Welcome roja and Your age is 4

20. Write a program to calculate the factorial of a number using input().

```
In [96]: number_fact = int(input("Enter a number for factorial"))
factorial=1

for i in range(number_fact):
    factorial*=i+1

print(f"Factorial of a given number {number_fact} is {factorial}")
```

Factorial of a given number 5 is 120

21. How do you prevent a user from entering an empty string?

```
In [110... string = input("Enter a string")
if not string:
    print(f"Given string is empty")
else:
    print(f"Given string is perfect square")
```

Given string is empty

22. Write a program to check if the entered number is a perfect square.

```
In [142... import math
num_sqrt = int(input("Enter a number"))
srt = math.sqrt(num_sqrt)
if srt*2 == num_sqrt:
    print(f"Given {num_sqrt} is perfect square of {srt}")
else:
    print(f"Given {num_sqrt} is not square")
```

Given 8 is not square

23. Write a program that asks the user for a year and determines if it's a leap year.

```
In [147... year = int(input("Enter a year"))

if year%4 == 0:
    print(f"Given year {year} is a leap year")
else:
    print(f"Given year {year} is not a leap year")
```

Given year 2026 is not a leap year

24. How can you remove leading and trailing spaces from a string input?

```
In [148... string = input("Enter a year").strip()

print(f"Given string is {string}")
```

Given string is sdas

25. How do you handle incorrect inputs when you expect an integer using input()?

```
In [157... try:
    number = int(input("Enter a integer"))
    print(f"Given number is {number}")
```

```
except:
    print(f"Given input is not an integer. Please enter an integer")
```

Given input is not an integer. Please enter an integer

26. Write a program that accepts a string and counts the occurrence of a particular character.

```
In [181... string = input("Enter a string")
dict_count = {}

for i in string:
    if i in dict_count:
        dict_count[i] = dict_count[i]+1
    else:
        dict_count[i] = 1

for k,v in dict_count.items():
    print(f"{k} in the given string {v}")
```

```
r in the given string 2
o in the given string 1
j in the given string 1
a in the given string 2
  in the given string 1
n in the given string 1
i in the given string 1
```

27. How would you convert user input to lowercase using `input()` ?

```
In [183... string = input("Enter a string").lower()

print(f"Given string {string}")
```

Given string roja

28. Write a program that accepts a number and prints whether it is a multiple of 10.

```
In [185... number = int(input("Enter a number"))

if number%10 == 0:
    print(f"Given number {number} is multiple of 10")
else:
    print(f"Given number {number} is not a multiple of 10")
```

Given number 45 is not a multiple of 10

29. How would you check if a string contains only alphabets using `input()` ?

```
In [191... string = input("Enter a string")

if string.isalpha():
    print(f"Given string {string} has alphabets")
else:
    print(f"Given string {string} not having alphabets")
```

Given string ldsf not having alphabets

30. Write a program to count the number of words in a sentence entered by the user.

```
In [201... sentence = input("Enter a Sentence").split()

print(f"Number of words in the given sentence is {len(sentence)}")
```

Number of words in the given sentence is 4

31. How would you accept a date input from the user in Python?

```
In [209... from datetime import datetime
date = input("Enter a date")

date = datetime.strptime(date, "%Y-%m-%d")
print("Entered date:", date)
```

Entered date: 2026-06-06 00:00:00

32. Write a program that checks if the entered number is divisible by both 3 and 5

```
In [218... number = int(input("Enter a number"))

if (number%3 == 0) & (number%5 == 0):
    print(f"Entered number {number} is divisible by 3 and 5")
else:
    print(f"Entered number {number} is not divisible by 3 and 5")
```

Entered number 18 is not divisible by 3 and 5

33. Write a program to swap the values of two variables using input()

```
In [219... var1 = input("Enter first variable")
var2 = input("Enter second variable")

print(f"First Variable Before Swap {var1}")
print(f"Second Variable Before Swap {var2}")

var2, var1 = var1, var2
print(f"First Variable After Swap {var1}")
print(f"Second Variable After Swap {var2}")
```

First Variable Before Swap roja
Second Variable Before Swap rani
First Variable After Swap rani
Second Variable After Swap roja

34. Write a program to take user input and print it without spaces between words

```
In [221... sentence = input("Enter your input")
print(f"Entered : {sentence.replace(' ', '')}")
```

Entered : Hiwelcome

35. How do you validate if an entered input is a valid email address?

```
In [225... import re
email = input("Enter your email")
if re.match(r"^[^@]+@[^@]+\.[^@]+", email):
    print("Valid email")
else:
    print("Invalid email")
```

Valid email

36. Write a program that accepts a number and prints its cube

```
In [228... var1 = int(input("Enter number"))
print(f"Cube of the given number {var1} is {var1**3}")
```

Cube of the given number 3 is 27

37. How would you accept and store multiple names from the user?

```
In [229... first_name = input("Enter your first name")
middle_name = input("Enter your middle name")
last_name = input("Enter your last name")

print(f"Welcome {first_name} {middle_name} {last_name}")
```

Welcome roja rani repalle

38. How would you extract numbers from a string entered by the user?

```
In [234... text = input("Enter a string")
numeric=[]

for i in text:
    if i.isnumeric():
```



```
numeric.append(i)
print(f"In the given string {i} is numeric")
```

In the given string 6 is numeric
In the given string 7 is numeric
In the given string 5 is numeric

39. How do you find the maximum number from a list of integers entered by the user?

```
In [239]: integers = [int(x) for x in input("Enter your integers with , separated").split()]
print(f"Max number in the given list {max(integers)}")
```

Max number in the given list 56

40. How would you prompt the user for input until they enter a valid number?

```
In [2]: while True:
    try:
        number = int(input("Enter a number"))
        print(number)
        break
    except:
        print("Enter a valid number")
```

Enter a valid number
23

41. Write a program to check if the entered string has digits

```
In [8]: import re
string = input("Enter a string")

if re.search(r'\d', string):
    print("Entered string is having a number")
else:
    print("Entered string is not having a number")
```

Entered string is having a number

42. Write a program to check if the entered string has only whitespace characters

```
In [12]: string = input("Enter a string")
if string.isspace():
    print("Enter string is space")
else:
    print("Enter string is not a space")
```

Enter string is space

43. Write a program to find the sum of all digits in a string entered by the user.

```
In [17]: string = input("Enter a string")
data = re.findall(r'\d', string)
sum_digits = 0
for num in data:
    sum_digits+=int(num)

print(f"Sum of digits in the string {sum_digits}")
```

Sum of digits in the string 13

44. Write a program that accepts a number and prints its absolute value.

```
In [20]: number = abs(int(input("Enter a number")))
print(f"Absolute value of the number is {number}")
```

Absolute value of the number is 89

45. How would you check if a string entered by the user contains any uppercase letters?

```
In [33]: string = input("Enter a string")
data = re.findall('[A-Z]', string)

if len(data):
    print(f"Having uppercases letters in the string {string}")
```

Having uppercases letters in the string ghtKL

46. Write a program that converts Celsius to Fahrenheit.

```
In [35]: temp = float(input("Enter Temperature in Celsius"))
temp_f = (temp*9/5) + 32
print(f"Temperature in Fahrenheit {temp_f}")
```

Temperature in Fahrenheit 37.76

47. Write a program to find the average of a list of numbers entered by the user.

```
In [41]: var = [int(x) for x in input("Enter numbers with space").split()]
print(f"Average of list of numbers {sum(var)/len(var)}")
```

Average of list of numbers 27.5

48. Write a program to count the number of consonants in a string entered by the user

```
In [49]: string = input("Enter a string").lower()
data = re.findall('[bcdfghjklmnpqrstvwxyz]', string)
print(f"Number of consonants in the given string {string} is {len(data)}")
```

Number of consonants in the given string roja rani is 4

49. How do you check if a string entered by the user contains any punctuation

```
In [56]: import string
text = input("Enter a string")
for char in text:
    if char in string.punctuation:
        print(f"String {text} is having punctuations")
        break
else:
    print(f"String {text} is not having punctuations")
```

String rj! python. is having punctuations

50. Write a program that accepts a sentence and prints the longest word.

```
In [60]: text = input("Enter a string").split()
length = 0
word = ''
for words in text:
    if length < len(words):
        length = len(words)
        word = words

print(f"Longest word is {word} with {length}")
```

Longest word is repalle with 7

In []: